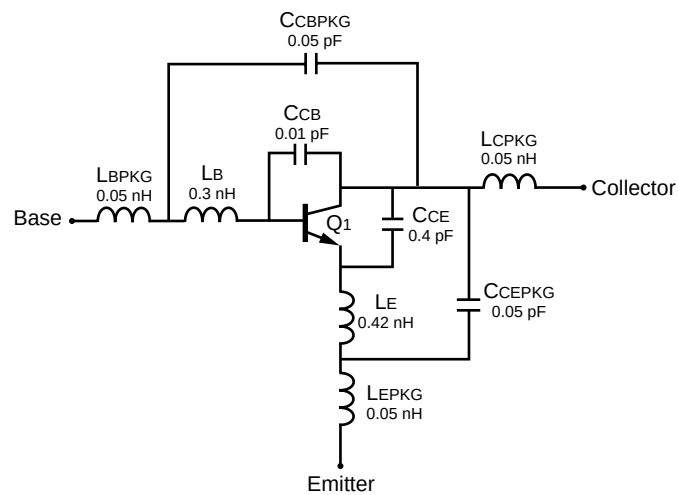


NEC

NONLINEAR MODEL

NE894M13

SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	137e-18	MJC	0.24
BF	129	XCJC	0.3
NF	0.9992	CJS	0
VAF	22.4	VJS	0.75
IKF	2.8	MJS	0
ISE	229e-15	FC	0.55
NE	2.5	TF	5e-12
BR	81.7	XTF	0.05
NR	0.9944	VTF	0.5
VAR	1.9	ITF	0.005
IKR	0.018	PTF	0
ISC	227e-18	TR	1.0e-9
NC	1.17	EG	1.11
RE	0.75	XTB	0
RB	5	XTI	3
RBM	3	KF	117e-15
IRB	0.005	AF	1.34
RC	6		
CJE	0.68e-12		
VJE	0.92		
MJE	0.26		
CJC	0.16e-12		
VJC	0.64		

ADDITIONAL PARAMETERS

Parameters	NE894M13
CCB	0.01 pF
CCE	0.4 pF
LB	0.3 nH
LE	0.42 nH
CCBPKG	0.05 pF
CCEPKG	0.05 pF
LBX	0.05 nH
LCX	0.05 nH
LEX	0.05 nH

MODEL TEST CONDITIONS
Frequency: 0.1 to 10 GHz
Bias: VCE = 0.5 V to 2 V, Ic = 0.5 mA to 20 mA
Date: 11/2001

(1) Gummel-Poon Model